

The Addressability Limit

An Overview

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Expository companion to the manuscript The Addressability Limit: A Geometric Capacity Bound for Information-Generating Hierarchies. Theorem numbers below refer to that manuscript; this document states results and the kernel of each argument, and proves nothing formally.

If a history is to stay distinguishable from every other one a process keeps, it must occupy a place no other occupies at the working resolution. A representation offers only finitely many such places within a given reach. *So unbounded memory inside a bounded radius is not hard—it is impossible*, and the exact size of the wall is the subject of this work. What follows is the argument in one pass, without proofs; the manuscript supplies those and machine-checks the algebra.

The claims, at a glance. No statement below is made more strongly than its row.

Result	Layer	Standing
The bound $\beta \leq c h_{\text{pack}}$; exact tightness; trichotomy; the balloon separation	I	Proven , curvature-free (algebra machine-checked)
Four-point $\Leftrightarrow$ tree; $n = 2$ embeddability floor	bridge	Proven (classical)
Zero relational tax in $\mathbb{H}_K^n$	host-specific	Proven ; single modern input
Full symmetry $\Rightarrow$ real hyperbolic host	IIa	Proven (Schur); the symmetry is <i>assumed</i>
The state equation (equality case)	IIa $\wedge$ IIb	Holds <i>under</i> saturation + isotropy; asserted of no system
Anything in nature saturates the ceiling	IIb	Open — the central empirical question

This is a *limit principle*, in the lineage of the second law, the Carnot bound, and the Shannon capacity: an inequality nothing beats, together with an idealized equality case that saturating systems approach. The old “geometric state equation” is that equality case, and it appears here only in that role.

1. The architecture of the limit

Three quantities, defined independently of one another, lock together into a ceiling.

The growth rate β is the exponential rate at which a process accumulates *retained*, mutually distinguishable histories— $\beta = \limsup_R R^{-1} \log |\mathcal{H}_R|$. “Retained” is load-bearing: a process that overwrites its past has $\beta = 0$ however fast it churns. (The bit-rate $h_{\text{eff}} = \beta / \ln 2$ is a unit conversion, not a physical claim; whether a given entropy estimator actually measures β is a separate empirical question.) *The radial rate* c converts one generative step into geometric reach: histories through depth R live inside a ball of radius $r(R)$, and $c = \limsup_R r(R)/R$. *The packing entropy* h_{pack} is the host’s room—the exponential growth rate of how many ε -separated points a ball of radius ρ can hold. It needs only a metric, so it speaks equally of trees, graphs, and manifolds.

The bound. $\beta \leq c h_{\text{pack}}$, unconditionally (Thm 4.1). *Why it holds.* The image of the histories is an ε -separated set inside a ball of radius $\approx cR$; a ball holds at most $e^{h_{\text{pack}} \cdot cR}$ such points. Counting is the whole proof.

Tightness. The bound is not a soft constraint but a *capacity*: block address capacity equals $c h_{\text{pack}}$ *exactly*, in every proper metric space (Thm 4.2). *Why it holds.* In a proper space the largest ε -separated set in a ball is the

packing number—the same extremal problem—so the ceiling is attained, not merely approached. Because nothing exceeds it and nothing prevents reaching it, h_{pack} *measures* retained diversity rather than bounding it.

The trichotomy. Rearranging the quantifiers turns the bound into a classification (Cor 4.3). A process that keeps its history must do exactly one of three things: occupy a host of *exponential* room ($h_{\text{pack}} > 0$); let its address radius outrun any finite rate ($c \rightarrow \infty$); or *forget* ($\beta = 0$). There is no fourth option. In a phrase: *remembering while creating requires exponential room*—or exponentially growing addresses, or forgetting.

Proving a host has *room* for addresses is only the first rung. Representing a genealogy—not just its endpoints—is a stronger demand, and the demands form a ladder: block (separated endpoints) $<$ persistent $<$ causal $<$ relational (the whole pairwise metric preserved). Packing alone secures only the bottom rung.

2. Room without shape

Packing entropy answers the addressing question completely and the structural one not at all. To see the gap, build *the balloon*: a ray of hub vertices, with an exponentially large clique of leaves hung on the hub at each radius. This space has strictly positive packing entropy at every resolution—many distinguishable addresses—yet it admits *no* uniform quasi-isometric image of a branching tree (Prop 3.1). *Why it holds.* Up to radius R the balloon has only $O(R)$ bounded-diameter shells, while a branching tree has exponentially many mutually tree-distant vertices at depth R ; a quasi-isometry can land only boundedly many of them per shell.

The balloon is room-rich and relationally empty. It is the exact proof that “diversity,” “packing,” and “preserved genealogy” are not synonyms: a process can have every address it needs and still fail to carry its own ancestry. (The same space is, verbatim, a star phylogeny after a rapid radiation—room spent, order never written—which the companion biology paper develops.)

3. The structural bridge

Whether a metric *is* a tree is decided, independently of any curvature, by the four-point condition: for every four points, the two larger of the three pairwise-sum pairings agree (Thm 5.1). Its residual δ measures departure from a tree. This classifies the qualitative object; it fixes no curvature magnitude.

One structural fact travels with it. A genuinely branching tree cannot live in one dimension—a point on a line has two neighbours, a branch point needs three—so $n = 2$ is the minimal smooth ambient dimension in which branching trees embed at low distortion (Rem 5.2). This is an *embeddability floor*: a fact about what a tree is, never a curvature fit or a per-family parameter. The four-point condition alone does not imply $n = 2$.

4. Closing the gap: the hyperbolic host

The balloon shows the relational rung can sit strictly below the block ceiling. Is there a host where the gap—the *relational tax*—vanishes? Real hyperbolic space is one, exactly. There the weighted relational capacity equals the block capacity, both $c(n-1)\sqrt{\kappa}$ (Thm 6.1): preserving a full genealogy costs *nothing beyond keeping its members apart*—at exponential order, as a supremum over weighted-clock tree codes (not endpoint attainment, and not transferable to an arbitrary sequence metric).

Why it holds. Take a cocompact hyperbolic lattice (critical exponent = volume entropy $(n-1)\sqrt{\kappa}$); pass to a finite-index subgroup with ε -separated orbit (residual finiteness); inside it, Skenderi’s 2026 theorem supplies a *free* sub-semigroup of nearly full exponent whose orbit is a quasi-isometrically embedded tree. Clocking each edge by its own hyperbolic length makes the radial budget exact, and the free-semigroup counting series hits growth rate $c(n-1)\sqrt{\kappa}$. Every other step is classical; this one contemporary result is the single external input the whole coding theorem rests on.

5. The isotropic host, and the curvature floor

Packing entropy is not yet a curvature. Turning one into the other takes a separate, named assumption of canonicity—**A3**: the host is complete, simply connected, unbounded, homogeneous and negatively curved, and the generating process carries *no* directional structure beyond a real inner product, i.e. full orthogonal ($O(d)$) symmetry.

Under A3 the host is forced: it can only be real hyperbolic space $\mathbb{H}_\kappa^n$ (Prop 7.2). *Why it holds.* Heintze’s classification writes a homogeneous negatively curved host as $N \rtimes \mathbb{R}$; an $O(d)$ -symmetric Lie bracket must vanish (the space of such brackets is zero— $\Lambda^2 \mathbb{R}^d \not\cong \mathbb{R}^d$, with the delicate $d = 3$ case killed by orientation-reversal), so N is abelian, and an $O(d)$ -symmetric contraction is a scalar by Schur. Abelian N with scalar contraction is exactly real hyperbolic space. A *departure* from A3 is then diagnostic rather than disappointing: it names *which* host one is in—reduce the symmetry to $U(n)$ and you get complex hyperbolic space, to $\mathrm{Sp}(n) \cdot \mathrm{Sp}(1)$ the quaternionic, and so on. Whether a real process satisfies A3, or breaks it in one of these named ways, is a sharp empirical question the manuscript states and does not run.

In this host the bound reads as a curvature *floor*: producing information at rate β forces $\kappa \geq (\beta/(c(n-1)))^2$ (Thm 7.3). Faster generation demands sharper curvature; there is a minimum the host cannot fall below.

6. Saturation, and the state equation

The floor is an inequality. Its equality case is *saturation*: a process whose retained history fills the available room, $\beta = c h_{\mathrm{pack}}$. Combine saturation with the isotropy of A3 and the floor becomes an exact equality, the *curvature–entropy state equation*,

$$\bar{\kappa}^* = \left(\frac{h_{\mathrm{eff}} \ln 2}{n-1} \right)^2 \quad (= (h_{\mathrm{eff}} \ln 2)^2 \text{ at } n = 2).$$

This is the whole of the old “state equation”—and, in the tradition of limit principles, we assert it of *no actual system*. It is the reversible-Carnot ideal: the least curvature that still holds the process losslessly, the economical realization a capacity-cost objective would select.

Two honesties travel with it. First, the boundary does not supply its own *mechanism*. The geometric mismatch $(\sqrt{\kappa} - \sqrt{\kappa^*})^2$ is a positive-definite potential with a unique zero, but that is *not* Lyapunov stability, global attraction, or self-organization; those would require an explicit evolution law and a proof the system descends it. A potential is not a dynamics. Second, testing whether anything saturates the bound demands β and h_{pack} measured from *independent* sources—never by fitting a curvature to the very entropy it is meant to confirm.

7. The firewall

A strict line separates the theorems from the world. The bound, the exact block identity, the hyperbolic relational capacity, and the polynomial-growth exclusion are *unconditional*—mathematical facts. Nothing in them requires that any physical process fill the capacity it proves exists, report a curvature, or behave as an optimal information engine. Any application to living genomes—the subject of the companion paper—is conducted under a separate evidentiary standard, and its data feed back into none of the theorems here.

8. Two frontiers

Mathematical. For which proper hosts is packing entropy a *complete* invariant for relational capacity? The balloon is the total-failure end, real hyperbolic space the perfect-success end; everything between—starting with the complex and quaternionic hyperbolic spaces—is open.

Empirical. Does anything in nature approach the ceiling? The bound proves the wall exists and the identity proves it reachable; neither shows any process reaches it. Whether biological evolution, or any

information-generating system, runs near ch_{pack} or orders of magnitude below is the fundamental unanswered question—and the one the whole apparatus is built to make honestly askable.